

Week 2 Design Retrospective

Change 1 — WeightedScoringStrategy + Strategy Selection [2 files modified]

Modified	Added	Untouched (key)
AssociativeFunction, DiagnosisEngine	WeightedScoringStrategy, DiagnosisStrategyFactory, ArgumentWeight, ArgumentWeightRepository	DiagnosisStrategy interface, all controllers, managers, factory

Layer: Engine layer routes per rule via *DiagnosisStrategyFactory*. Domain layer extended with *threshold* + *argumentWeights* on *AssociativeFunction*.

⚠ Unexpected: *AssociativeFunction* — natural owner of rule-level weights but not expected to change.

→ Redesign: Introduce a separate *WeightedRuleConfig* entity (1-to-1 with *AssociativeFunction*) so the domain object stays untouched.

Change 2 — Decorator Pipeline (3 Decorators) [2 files modified]

Modified	Added	Untouched (key)
ObservationManager, PhenomenonType.java (+ Observation*, shared with C4)	ObservationProcessor, ObservationProcessorDecorator, BaseObservationProcessor, UnitValidationDecorator, AnomalyFlaggingDecorator, AuditStampingDecorator	ObservationFactory, all controllers, repositories, DiagnosisEngine, event listeners

Layer: Manager layer wires chain via private *buildPipeline()*. Decorator package entirely new and self-contained.

⚠ Unexpected: *PhenomenonType/Observation* — field additions (*normalMin*, *normalMax*, *anomalyFlag*) needed for decorator persistence.

→ Redesign: Store validation config in a separate *ObservationSpec* value object; decorators read it without touching the domain entity.

Change 3 — Multi-User Support + Command Undo [5 files modified]

Modified	Added	Untouched (key)
Command, BaseCommand, CommandLogEntry, CommandLog, AppConfig	UserRole, AppUser, AppUserRepository, CurrentUser, UserInterceptor, UserController, UndoService, UndoController	ObservationManager, PatientManager, all observation/patient controllers, factory, strategies

Multi-layer rule (3 layers crossed): Config (AppConfig + UserInterceptor), Command bus (Command/BaseCommand/CommandLog*), Web (UserController/UndoController). Cross-cutting user identity required all three layers simultaneously — a pre-existing auth abstraction would have contained propagation to one layer.

⚠ Unexpected: *AppConfig* — needed only to register the interceptor bean.

→ Redesign: Have *UserInterceptor* implement *WebMvcConfigurer* directly so it self-registers; *AppConfig* stays untouched.

⚠ Unexpected: *Command* interface / *BaseCommand* — *undo()* propagated interface-wide.

→ Redesign: Add a default no-op *undo()* on *BaseCommand*; concrete commands opt in, avoiding interface-wide propagation.

Change 4 — Concept Hierarchy + PropagationListener [5 files modified]

Modified	Added	Untouched (key)
Observation* (shared), Phenomenon, ObservationFactory, SimpleConjunctiveStrategy, PhenomenonRepository	ObservationSource, PropagationListener	All controllers, managers, DiagnosisEngine, AuditLogListener, RuleEvaluationListener, event publisher

Layer: Event layer: *PropagationListener* plugs into existing Spring event bus — zero changes to existing listeners. Factory gained overloads so existing callers unchanged.

⚠ Unexpected: *SimpleConjunctiveStrategy* — required MANUAL-only filter to prevent circular inference loops from propagated observations.

→ Redesign: Encode observation provenance on the *Strategy* interface (e.g., an *evaluationContext* flag) so each strategy self-filters without ad-hoc checks. *WeightedScoringStrategy* was written with this filter from the start.

Summary — Pre-Existing Files Modified Per Change

Change	New Files Added	Modified Pre-existing	#
C1: <i>WeightedScoringStrategy</i>	4 new files	<i>AssociativeFunction</i> , <i>DiagnosisEngine</i>	2
C2: Decorator Pipeline	6 new files	<i>ObservationManager</i> , <i>PhenomenonType.java</i> (+ 1 shared*)	2
C3: Multi-user + Undo ⚠	8 new files	Command , BaseCommand , CommandLogEntry , CommandLog , AppConfig	5
C4: Hierarchy + Propagation	2 new files	<i>Phenomenon</i> , <i>ObservationFactory</i> , <i>SimpleConjunctiveStrategy</i> , <i>PhenomenonRepository</i> <i>Observation*</i> (shared*)	5
TOTAL unique files modified	20 new files total	* Observation shared C2+C4; counted once	14

Conclusion

14 unique pre-existing files were modified across 4 changes; 20 new files were added. C3 (Multi-User + Undo) was the highest-impact change, crossing three architectural layers due to the cross-cutting nature of user identity — a pre-existing auth abstraction would have contained it to one. All unexpected touches have concrete single-step redesigns.